

HUJI Research Monitor

Weekly Digest · Week 28 · July 06, 2026

EXECUTIVE SUMMARY

Hebrew University researchers are driving innovation across several high-impact sectors, poised to attract significant investor and industry interest. Notable advancements include novel cell and gene therapies targeting chronic autoimmune diseases and demyelinating neurological disorders, representing major shifts in treatment paradigms. The digest also highlights breakthroughs in precision diagnostics and therapeutics for oncology, alongside promising medical devices for infection control and oral health, and cutting-edge AI tools for accelerating drug discovery. These technologies address critical unmet needs and offer scalable solutions in large, high-value markets.

#1

B cell Maturation Antigen Targeted Chimeric Antigen Receptor T Cell Therapy for Refractory Systemic Lupus Erythematosus and Systemic Sclerosis.

37/50

Main Researcher: Chamutal Gur

[Immunology] [Clinical] [Synthetic Biology]

"New CAR T-cell therapy shows promise in severe autoimmune diseases like lupus and systemic sclerosis."

WHY NOW

This represents a significant pivot for CAR T technology from oncology to large, chronic autoimmune markets, offering a high-value therapeutic shift. Early phase data suggests a robust clinical benefit.

COMMERCIAL ANGLE

Developing specialized CAR T-cell therapies for autoimmune indications represents a massive shift from oncology toward high-value chronic disease management markets.

<https://europepmc.org/article/MED/42098533>

#2

Pro-repair properties of a human embryonic stem cell-derived astrocyte cell therapy in demyelinating disorders.

36/50

Main Researcher: Tamir Ben-Hur

[Neuroscience] [Immunology] [Clinical] [Synthetic Biology]

"Off-the-shelf stem cell therapy developed for remyelination, addressing multiple sclerosis and other demyelinating disorders."

WHY NOW

This scalable cell therapy offers a potential breakthrough for demyelinating neurological disorders, a field with high unmet clinical needs and significant market demand for regenerative treatments.

COMMERCIAL ANGLE

Development of a scalable, off-the-shelf cell therapy for remyelination in Multiple Sclerosis and other demyelinating neurological disorders.

<https://pubmed.ncbi.nlm.nih.gov/42276062/>

#3

33/50

The effect of a proprietary chlorine-stabilizing surface-coating formulation on healthcare environment disinfection in multidrug-resistant bacteria isolation rooms.

Main Researcher: Oren Zimhony

[Medical Device] [Materials] [Clinical]

"Proprietary surface coating dramatically enhances hospital disinfection against multidrug-resistant bacteria, reducing infection transmission."

WHY NOW

Healthcare facilities urgently need better infection control, and this durable, scalable antimicrobial coating offers a practical solution to combat rising multidrug-resistant pathogens. It directly addresses a critical public health challenge.

COMMERCIAL ANGLE

This technology offers a scalable solution for hospitals and long-term care facilities to enhance infection control protocols and reduce the transmission of multidrug-resistant organisms through high-durability antimicrobial coatings.

<https://pubmed.ncbi.nlm.nih.gov/42374536/>

#4

33/50

RNA dicing promotes the expression of an oncogenic JAK1 isoform.

Main Researcher: Yuval Malka

[Drug Discovery] [Diagnostics] [Genomics] [Immunology]

"Discovery of RNA dicing enables new precision medicine approach for stratifying cancer patients for JAK1 inhibitors."

WHY NOW

This offers a novel companion diagnostic strategy to improve patient selection for existing oncology drugs, expanding market utility and treatment efficacy, particularly for endometrial tumors.

COMMERCIAL ANGLE

The discovery enables a new precision medicine approach by using RNA dicing profiles to stratify patients for existing JAK1 inhibitors like momelotinib.

<https://pubmed.ncbi.nlm.nih.gov/41984590/>

#5

33/50

Recombinant Human Amelogenin Protein Enhances Healing of Osteochondral Injury in a Goat Model.

Main Researcher: Amir Haze

[Medical Device] [Clinical] [Synthetic Biology] [Proteomics]

"Recombinant human amelogenin protein (Remelix®) demonstrates significant healing of cartilage and bone injuries."

WHY NOW

This regenerative medicine biologic targets the massive and growing market for osteoarthritis and traumatic joint injury treatments, where current options are often limited or invasive.

COMMERCIAL ANGLE

Development of a high-value regenerative medicine biologic for treating osteoarthritis and traumatic joint injuries in humans.

<https://pubmed.ncbi.nlm.nih.gov/41925697/>

#6

APOBEC mutational signatures predict immune checkpoint inhibitor benefit in TMB-low metastatic NSCLC independent of PD-L1 status.

32/50

Main Researcher: N Peled

[Diagnostics] [Genomics] [Immunology] [Software/AI]

"APOBEC mutational signatures predict immune checkpoint inhibitor benefit in TMB-low lung cancer patients."

WHY NOW

This companion diagnostic could unlock immunotherapy benefits for patients currently excluded, significantly expanding the market for blockbuster immune checkpoint inhibitors and improving patient outcomes.

COMMERCIAL ANGLE

Development of a companion diagnostic test using machine learning to identify APOBEC signatures in TMB-low lung cancer patients to expand the market for immune checkpoint inhibitors.

<https://pubmed.ncbi.nlm.nih.gov/42378718/>

#7

Targeting primary and metastatic ovarian cancer with a peptide derived from the human NAF-1/CISD2 protein.

29/50

Main Researcher: Rachel Nechushtai

[Drug Discovery] [Clinical] [Immunology]

"Novel peptide targeting NAF-1/CISD2 shows strong potential against primary and metastatic ovarian cancer."

WHY NOW

Ovarian cancer remains a challenging malignancy with high recurrence rates, making this first-in-class peptide a high-potential therapeutic candidate for pharmaceutical licensing in oncology.

COMMERCIAL ANGLE

The peptide represents a high-potential therapeutic candidate for a first-in-class oncology drug, suitable for licensing to pharmaceutical companies focusing on precision cancer therapies or protein-protein interaction inhibitors.

<https://pubmed.ncbi.nlm.nih.gov/42259134/>

#8

Breath-Based Detection of Oral Diseases Using Sensors and Machine Learning.

28/50

Main Researcher: Yael Hour-Haddad

[Medical Device] [Diagnostics] [Software/AI]

"Breath-based detection system uses sensors and AI for rapid, non-invasive diagnosis of oral diseases."

WHY NOW

A handheld breathalyzer provides an objective, patient-friendly diagnostic tool for dental clinics, revolutionizing periodontal health screening and offering significant market potential in preventive care.

COMMERCIAL ANGLE

Development of a non-invasive, rapid-screening handheld breathalyzer for dental clinics to provide objective diagnostics for periodontal health.

<https://pubmed.ncbi.nlm.nih.gov/42379259/>

#9

Targeting Temozolomide-Resistant Glioblastoma: Therapeutic Potential of Neuronal Nitric Oxide Synthase Inhibitor.

26/50

Main Researcher: Haitham Amal

[Drug Discovery] [Neuroscience] [Clinical]

"Nitric oxide synthase inhibitor overcomes temozolomide resistance in glioblastoma, boosting chemotherapy efficacy."

WHY NOW

Glioblastoma has a dismal prognosis with high drug resistance; this adjuvant therapy can extend the clinical utility of standard chemotherapy, addressing a critical unmet need for brain cancer patients.

COMMERCIAL ANGLE

Development of a high-value adjuvant therapy to extend the clinical utility of existing standard-of-care chemotherapy for resistant brain tumors.

<https://pubmed.ncbi.nlm.nih.gov/42374654/>

#10

Strategic Template Filtering Accelerates Fragment-Based Peptide Docking.

26/50

Main Researcher: Ora Schueler-Furman

[Drug Discovery] [Software/AI] [Proteomics]

"AI-powered algorithm accelerates fragment-based peptide docking for efficient drug discovery and design."

WHY NOW

Speeding up peptide-based drug design reduces R&D costs and time, making this high-efficiency algorithm highly valuable for integration into modern AI-driven drug discovery platforms and pharmaceutical pipelines.

COMMERCIAL ANGLE

Licensing or integrating this high-efficiency docking algorithm into AI-driven drug discovery platforms to accelerate the design of peptide-based therapeutics.

<https://pubmed.ncbi.nlm.nih.gov/42299509/>

#11

IR nanospectroscopy mapping of facet-dependent sulfur poisoning and thermal regeneration on platinum nanocrystals.

26/50

Main Researcher: Elad Gross

[Materials] [Clean Energy]

"Advanced nanospectroscopy maps sulfur poisoning on platinum catalysts, enabling more durable fuel cells."

WHY NOW

Optimizing catalyst durability and efficiency is crucial for advancing clean energy technologies like fuel cells and for numerous industrial chemical processes, promising significant environmental and economic impact.

COMMERCIAL ANGLE

The ability to map surface-level poisoning provides a roadmap for engineering more durable and efficient catalysts for fuel cells and industrial chemical processes.

<https://pubmed.ncbi.nlm.nih.gov/42065428/>

#12

26/50

Safety and Efficacy of Direct Selective Laser Trabeculoplasty in Patients With Ocular Hypertension or Open Angle Glaucoma.

Main Researcher: David Zadok

[Medical Device] [Clinical]

"Direct Selective Laser Trabeculoplasty is safe and effective as a primary treatment for glaucoma."

WHY NOW

This laser-based ophthalmic procedure offers a compelling first-line alternative to daily eye drops for glaucoma, potentially expanding the market for surgical devices and improving patient adherence.

COMMERCIAL ANGLE

The findings support the commercialization and expanded clinical adoption of laser-based ophthalmic surgical devices as a first-line alternative to traditional glaucoma eye drops.

<https://europepmc.org/article/MED/42148860>